

Utilizing MobileNetV2 for Enhanced Blood Cancer Detection

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Abstract— A growing number of people are dying every day as a result of acute lymphoblastic leukaemia. Those that survive suffer from a loss of vitality. The different shapes, patterns, and textures of ALL blast cells present a difficulty for the current artificial intelligence-based automated solutions. Around 1% of all blood cells are leukocytes, which are created in the bone marrow. The uncontrolled expansion of blood cells is one of the causes of blood cancer. In past times, the process was manually completed by a knowledgeable expert over the duration of a long amount of time. The proposed approach utilized a comprehensive dataset comprising high-quality blood cell images, encompassing various subtypes of blood cancer, to train and validate the MobileNetV2 architecture.

Keywords— *deep neural network (DNN), melanoma, Blood cells, MobileNetV2.*

1. INTRODUCTION

Blood cancer is primarily brought on by the growth of aberrant blood cells, which is prevented by the rapid proliferation of healthy blood cells. Leukaemia, myeloma, and lymphoma are the three forms of blood cancers. A type of white blood cell malignancy known as acute lymphocytic leukaemia (ALL) damages the bone marrow. The word "acute" relates to the disease's quick progression and the fact that, if untreated, it can quickly prove fatal [1] [2] [3].

Since their introduction, magnetic resonance imaging (MRI) devices have demonstrated potential as supplementary diagnostic tools. In the early detection of blood cancer, MRI systems that use computer technology to detect medical imaging anomalies and use these findings for radiologists' diagnoses may be crucial. It is possible to recognize MRI as an efficient diagnostic tool for blood cancer since it combines two computer science technologies for diagnostic imaging, image processing, pattern recognition, and artificial intelligence. In the current work, a brand-new, quick, and automatic CAD system is suggested for the detection of blood cancer. A broad overview of the physiological characteristics of the blood, blood illnesses, frequently used diagnostic techniques, etc. is presented before talking about the work that was done.

Myeloma Multiforme is an undeveloped tumours of plasma cells that aids in getting rid of infection [4]. It is three times more frequent than ALL. In contrast to leukaemia, this disorder does not produce a rise in blood counts, but the infected person experiences anaemia (low red blood cell count) as a result of the dangerous cells occupying the RBCs' space. MM is the cause of thrombocytopenia, or a decreased number of platelets in the blood [5]. It also causes bone erosion, which is detected by CT scans [6]. In 2019, over 1500 persons died as a result of this condition, accounting for 0.2% of all cancer-related fatalities [7]. Therapy for cancer of the blood varies by the patient's age, kind, how quickly the disease spreads to other contaminated regions, and other factors [8]. If carried out by individual, it performs 100% accuracy [9]. Automatic counting, on the opposite hand, is a quicker procedure but has a larger chance of error. As a result, both strategies offer both benefits and drawbacks. This practice provides a review of the controlled method for determining the kind of WBC. The automatic grouping system is cheap and may be deployed rapidly in all regions. The proposed method introduces inconsistencies caused by manual sorting work, the need for a capable expert, inaccuracies caused by cells being indistinguishable when investigated under a microscope, and more.

Block representations of the major forms of leukaemia and blood components are depicted in Fig. 1 Leukaemia can be identified manually by pathologists and haematologists looking at the patient's blood. They identify and categorize leukaemia using structural characteristics and cell counts, and they require sophisticated laboratory apparatus to identify the specific form of leukaemia. This method of leukaemia diagnosis and categorization is expensive and time-consuming. Additionally, the expertise and experience of subject-matter experts in this area affect how accurately a diagnosis and classification are made.

Deep learning techniques are got closer, which gather desired features from raw data, can assist in dealing with all of the above issues [10] [11]. Convolutional Neural Networks (CNNs) use different multi-layer perceptron's to produce effective results with little pre-processing [12].

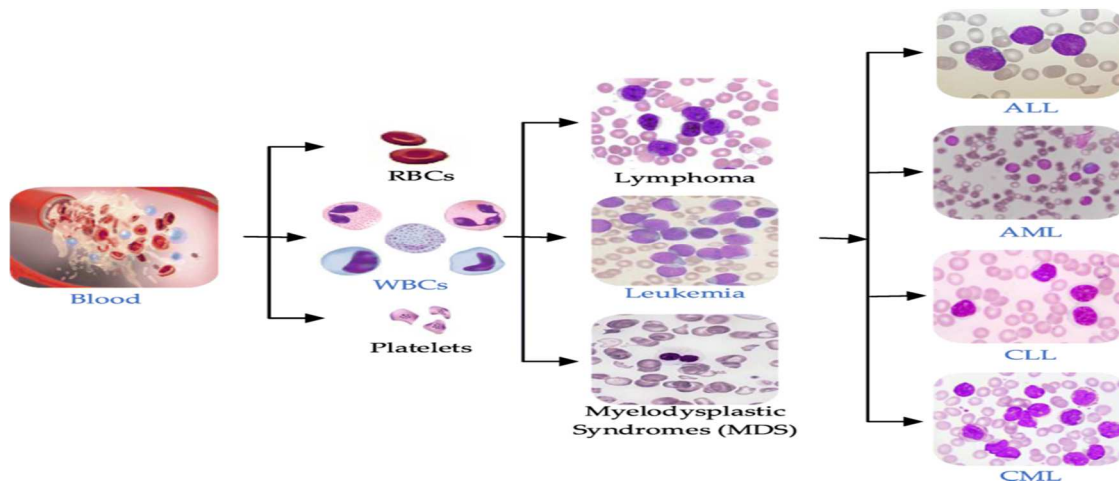


FIGURE. 1 BLOOD COMPONENTS AND MAIN TYPES OF LEUKAEMIA

Since each convolutional layer of a CNN extracts new feature from different images and establishes a activation function, CNNs behave as feature extractors. MM is automated classification described in the proposed study which used Convolution Neural Networks. As a result, the paper investigates the suggested deep learning model's performanceutilizing accuracy, precision, recall, sensitivity, and selectivity as equivalent measures.

2. RELATED WORK

T. Beer et al (2020) [13] discovered three multicancer screening tests—the Cancer SEEK assay, the Galleri test, and the PanSeer assay that all exhibit high specificity in early results is being validated in ongoing clinical trials. Multicancer screening technologies offer a larger potential to enable broad screening in the general population, despite the fact that various single-cancer detection assays are now in development.

Yanwu Chu et al (2020) [14] described two models one is KNN with accuracy 88.14% and second one is LDA with accuracy of 94.45% and comparative results with RSM model with both KNN and LDA which improved accuracy from 94.45% to 98.34%.

Soichiro Fukuyama et al (2021) [15] using 1 mL of blood samples spiked with various cancer cell concentrations, the fabrication device's ability to detect cancer cells was assessed., Additionally, when compared to existing cancer detection tools (such as the CellSearch system and Veridex), the findings of the cancer detection experiment showed that the constructed cancer detection device could perform cancer detection with a better degree of accuracy.

Yanwu Chu et al (2020) [16][17] used LIBS spectrum collection, the serum samples were leaked onto a boric acid. These findings demonstrate that the RSM-LDA model, when combined with LIBS, can distinguish between blood cancer and health control, as well as identify distinct types of blood cancer.

In a preliminary study published in 2020, N. Shamyuktha and Dr. S. Preetha [18]concentrated on the creation of a system that would use microscopic images of blood samples to identify various forms of leukaemia. White blood cells, which are essential to the immune system of the body, abnormally proliferate in leukaemia, a medical disorder. In order to effectively treat and manage this condition, early and precise diagnosis of leukaemia is a crucial first step.

3. DATASET DESCRIPTION

The collection of images related to acute lymphoblastic leukaemia (ALL) was used. The bone marrow laboratory at Tehran, Iran's Taleqani Hospital is where the photographs in this collection [19, 20] were produced. This dataset includes 3256 PBS images from 89 individuals suspected of having ALL, whose blood samples had been professionally processed and stained by skilled laboratory professionals. 25 healthy individuals with benign diagnosis and 64 patients with Early Pre-B, Pre-B, and Pro-B ALL subtypes of ALL have been included in this group of patients.

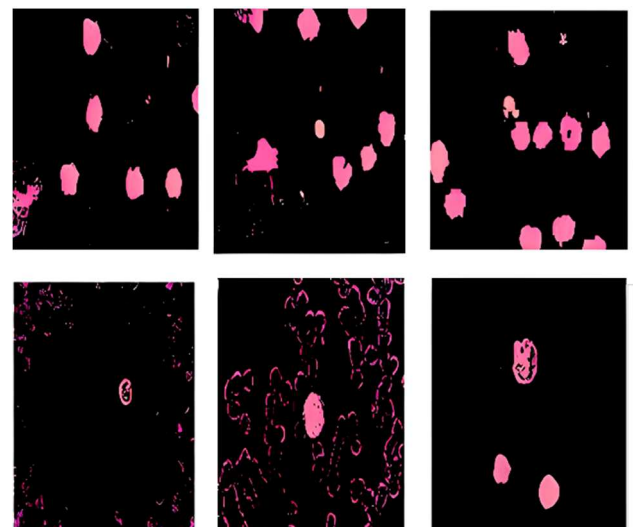


Figure 2. different pattern of WBCs.

The classes of this dataset are benign and malignant. The first category is hematogenous, while the second category is ALL, which encompasses the three malignant lymphoblast subtypes Early Pre-B, Pro-B ALL and Pre-B in figure 2.

Further Dataset provides segmented images with segmentation based on colour thresholding in the HSV colour space. Images of early, pre, pro and benign cells are shown in Figures 3.

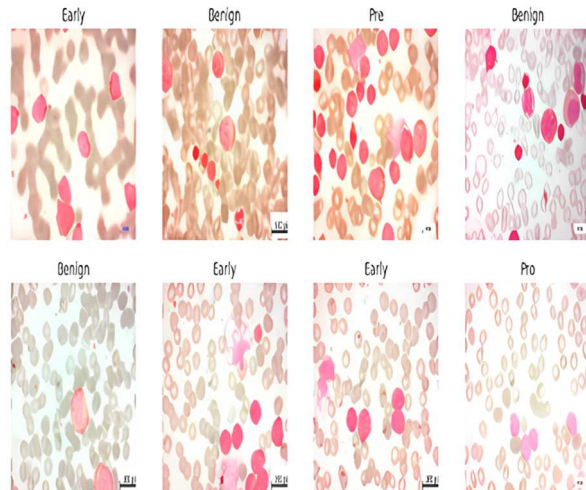


Figure 3: Sample images of sub categories

4. PROPOSED METHODOLOGY

The proposed approach for the identification and categorization of leukaemia is depicted by trimming the black margins of the photos, the input dataset is first processed. The images are then adjusted to reduce the influence of large input value ranges, which might result in vanishing gradients, allowing training to converge faster. Models VGG16, DenseNet121, MobileNetV2, were used to extract features. The best feature extraction model was MobileNetV2.

4.1 Input Dataset

In order to extract the deep learnt features, MobileNetV2[21] is applied on the leukemia dataset. We created an 81-layer MobileNetV2 architecture. There are 13 Conv2D layers, 4 for zero padding levels, 26 for relu activation function, 24 for batch normalization, 12 for depth-wise convolutional layers, 1 for flattening, and finally a Dense layer with 4 units and SoftMax activation function. There are 3,429,572 total parameters, 200,708 of which are trainable, and 3,228,864 of which are not.

4.2 Pre-Processing

The foremost phase in image processing, as emphasized in reference [22], is pre-processing, primarily enhancing the quality of leukaemia images. This step is important for reducing the presence of noise and undesirable information within the images. Data pre-processing encompasses two key steps: normalization and the removal of black edges through cropping.

4.3 Normalization

Normalization is a fundamental data preparation technique that is used to standardize numerical features within a dataset so that they are comparable in terms of scales and ranges. This is an important step since many machine learning algorithms work best when the input data has been normalized. Distance-based algorithms, such as K-Nearest Neighbours (KNN) or Support Vector Machines (SVM), for example, are extremely sensitive to the size of input features.

Several methods exist for normalizing data, with min-max scaling being one of the most commonly employed. Min-max scaling ensures that the numerical features are rescaled to have a minimum value of 0 and a maximum value of 1, as demonstrated in references [23] and [24].

Normalization is often done to train data and the same grading factors are then applied to test data. This phase is critical for keeping deep learning (DL) models numerically stable. It not only speeds up learning but also improves the stability of gradient descent optimization by 255.

In the context of image processing, pixel values perform optimally when the input data is normalized. Achieving a pixel value range of 0 to 1 is commonly accomplished by dividing the pixel effectively process the image data.

4.4 Feature Extraction Using MobileNetV2

The conventional convolution process in the convolutional neural network (CNN) [25] architecture known as MobileNetV2 consists of pointwise and depth wise convolutions. The filter is applied independently to input values in depth wise convolution, which reduces computing cost and corresponding output is then combined by pointwise convolutions in a 1x1 convolution, enabling deeper feature interactions. The bottleneck layer's number of channels is increased and a depth wise separable convolution is applied by the model's inverted residuals with linear bottlenecks. This model is more effective because to all of these qualities.

5. RESULTS AND DISCUSSION

The Training, Validation, and Testing Accuracy for the Optimized MobileNetV2 Model are 0.9412, 0.9899, and 0.9854, respectively. Loss in Testing, Validation, and Training are, respectively, 0.6753, 0.1591, and 0.2351. Outcome of the Model for MobileNetV2. The training accuracy against validation accuracy is displayed in Table 1.

TABLE 1: RESULT ANALYSIS OF MOBILE NET v2

	Accuracy	Loss
Training	0.9412	0.6763
Validation	0.9899	0.1591
Testing	0.9854	0.2351

The proposed methodology's architecture is depicted in Figure 4. Features are chosen, data is resampled, split, a model is trained, and a model is evaluated

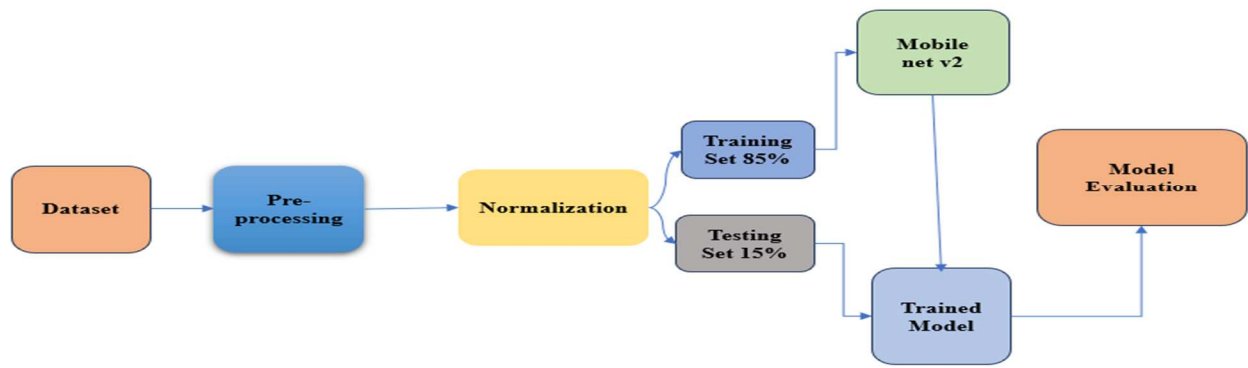


FIGURE 4: PROPOSED MODEL

On the basis of table 1 we have created a graph for accuracy and loss using the proposed Mobile Net v2 model which is increasing the accuracy by 98.5% and loss is decreasing by 2.35%.

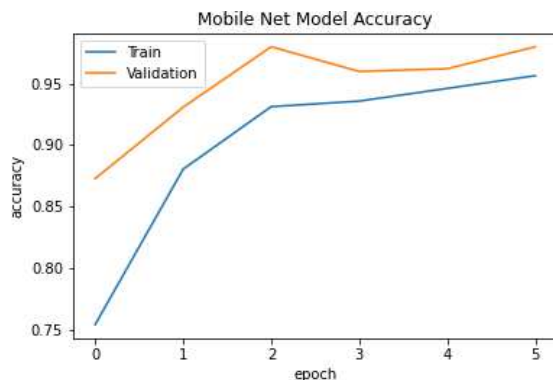


FIGURE 5 MOBILE NET MODEL ACCURACY

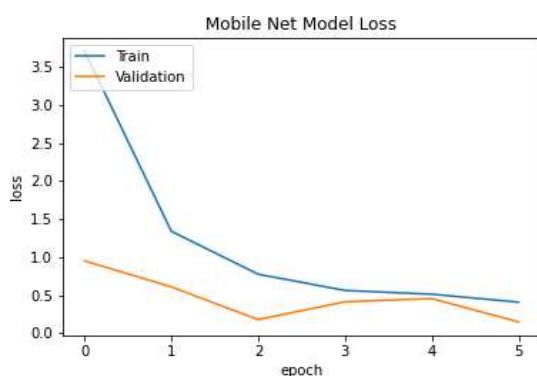


FIGURE 6. MOBILE NET MODEL LOSS

Analysing the above results, we can conclude that ResNet50 model provide lower outcome as compared to others and MobileNetV2 model accomplished better results.

6. CONCLUSION

In conclusion, this study suggested a unique method for precisely analysing and classifying leukaemia cells in children by combining image processing methods with attention mechanism-based ML. The suggested Mobile Net v2 algorithm for blood cancer identification in humans employs techniques to achieve better sensitivity and accuracy. It provides a potential strategy for precise and effective identification, which may help to diagnosis and treatment of leukaemia.

Further due to lack of strategy and generalized model there is need to develop a robust technique for segmentation, estimated blast cell calculation, and generalized classification.

REFERENCES

- [1] K. Kessenbrock, V. Plaks, and Z. Werb, "Matrix metalloproteinases: Regulators of the tumor microenvironment," *Cell*, vol. 141, no. 1, pp. 52–67, Apr. 2010.
- [2] A. Rehman, N. Abbas, T. Saba, S. I. U. Rahman, Z. Mehmood, and H. Kolivand, "Classification of acutelymphoblastic leukemia using deep learning," *Microsc. Res. Techn.*, vol. 81, no. 11, pp. 1310–1317, Nov. 2018.
- [3] S. Shafique and S. Tehsin, "Acute lymphoblastic leukemia detection and classification of its subtypes using pretrained deep convolutional neural networks," *Technol Cancer Res. Treatment*, vol. 17, Jan. 2018, Art. no. 153303381880278.
- [4] M. Hallek, P. Leif Bergsagel, and K. C. Anderson, "Multiple myeloma: Increasing evidence for a multistep transformation process," *Blood*, vol. 91, no. 1, pp. 3–21, Jan. 1998.
- [5] J. G. Kelton, A. R. Giles, P. B. Neame, P. Powers, N. Hageman, and J. Hirsch, "Comparison of two direct assays for platelet-associated IGG (Paigg) in assesment of immune and nonimmune thrombocytopenia," *Blood*, vol. 55, no. 3, pp. 424–429, 1980. [Online]. Available: <https://ashpublications.org/blood/article/55/3/424/161619/Comparison-of-two-direct-assays-for-platelet>.
- [6] M. Perkonig, J. Hofmanninger, B. Menze, M.-A. Weber, and G. Langs, "Detecting bone lesions in multiplemyeloma patients using transfer learning," in *Data Driven Treatment Response Assessment and Preterm, Perinatal, and Paediatric Image Analysis*. Cham, Switzerland: Springer, 2018, pp. 22–30.
- [7] A. Jemal, R. Siegel, E. Ward, Y. Hao, J. Xu, and M. J. Thun, "Cancer statistics, 2009," *CA, A Cancer J. Clin.*, vol. 59, no. 4, pp. 225–249, Jul. 2009.
- [8] Y. Liu and F. Long, "Acute lymphoblastic leukemia cells image analysis with deep bagging ensemble learning," in *CNMC Challenge: Classification in Cancer Cell Imaging*. Singapore: Springer, 2019, pp. 113–121.

- [9] A. B. Kul'chyn's'kyi, V. M. Kyjenko, W. Zukow, and I. L. Popovych, "Causal neuro-immune relationships at patients with chronic pyelonephritis and cholecystitis. Correlations between parameters EEG, HRV and white blood cell count," *Open Med.*, vol. 12, no. 1, pp. 201–213, Jul. 2017.
- [10] S. Kant, "Leukonet: Dct-based cnn architecture for the classification of normal versus leukemic blasts in b-all cancer," 2018, arXiv:1810.07961. [Online]. Available: <https://arxiv.org/abs/1810.07961>
- [11] I. Arel, D. C. Rose, and T. P. Karnowski, "Deep machine learning—A new frontier in artificial intelligence research," *IEEE Comput. Intell. Mag.*, vol. 5, no. 4, pp. 13–18, Nov. 2010.
- [12] J. Zhao, M. Zhang, Z. Zhou, J. Chu, and F. Cao, "Automatic detection and classification of leukocytes using convolutional neural networks," *Med. Biol. Eng. Comput.*, vol. 55, no. 8, pp. 1287–1301, Aug. 2017.
- [13] Beer, Tomasz M.. "Novel blood-based early cancer detection: diagnostics in development." *The American journal of managed care* 26 14 Suppl (2020): S292-S299 .
- [14] YanWu Chu, Feng Chen, Ziqian Sheng, Deng Zhang, Siyu Zhang, Weiliang Wang, Honglin Jin, Jianwei Qi, and LianBo Guo, "Blood cancer diagnosis using ensemble learning based on a random subspace method in laser-induced breakdown spectroscopy," *Biomed. Opt. Express* 11, 4191-4202 (2020)
- [15] Soichiro Fukuyama, Seitaro Kumamoto, Seiya Nagano, Shoma Hitotsuya, Keiichiro Yasuda, Yusuke Kitamura, Masaaki Iwatsuki, Hideo Baba, Toshihiro Ihara, Yoshitaka Nakanishi, Yuta Nakashima,
- [16] Detection of cancer cells in whole blood using a dynamic deformable microfilter and a nucleic acid aptamer, *Talanta*, Volume 228, 2021, 122239, ISSN 0039-9140, <https://doi.org/10.1016/j.talanta.2021.12223>
- [17] YanWu Chu, Feng Chen, Ziqian Sheng, Deng Zhang, Siyu Zhang, Weiliang Wang, Honglin Jin, Jianwei Qi, and LianBo Guo, "Blood cancer diagnosis using ensemble learning based on a random subspace method in laser-induced breakdown spectroscopy," *Biomed. Opt. Express* 11, 4191-4202 (2020)
- [18] Preetha, Dr.S. and N Shamyuktha.. "DETECTION OF CANCER IN HUMAN BLOOD SAMPLE." (2020).
- [19] Kaur, S.; Gupta, S.; Singh, S.; Gupta, I. Detection of Alzheimer's Disease Using Deep Convolutional Neural Network. *Int. J. Image Graph.* 2021, 22, 2140012. [CrossRef]
- [20] Kaur, S.; Gupta, S.; Singh, S.; Hoang, V.T.; Almakdi, S.; Alelyani, T.; Shaikh, A. Transfer Learning-Based Automatic Hurricane Damage Detection Using Satellite Images. *Electronics* 2022, 11, 1448. [CrossRef]
- [21] L. Zhang, L. Lu, I. Nogues, R. M. Summers, S. Liu, and J. Yao, "DeepPap: Deep convolutional networks for cervical cell classification," *IEEE J. Biomed. Health Informat.*, vol. 21, no. 6, pp. 1633–1643, Nov. 2017.
- [22] R. Duggal, A. Gupta, R. Gupta, and P. Mallick, "Sd- layer: Stain deconvolutional layer for CNNs in medical microscopic imaging," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent. Cham, Switzerland: Springer*, 2017, pp. 435–443
- [23] Gupta, A.; Gupta, R. ALL Challenge Dataset of ISBI 2019 [Data Set]. *Cancer Imaging Arch.* 2019. Available online: <https://wiki.cancerimagingarchive.net/pages/viewpage.action?pageId=52758223> (accessed on 15 May 2023). [CrossRef]
- [24] Kaur, S.; Gupta, S.; Singh, S.; Koundal, D.; Zaguia, A. Convolutional neural network-based hurricane damage detection using satellite images. *Soft Comput.* 2022, 26, 7831–7845.
- [25] Verma, Ekansh & Singh, Vijendra. (2019). ISBI Challenge 2019: Convolution Neural Networks for B-ALL Cell Classification. 10.1007/978-981-15-0798-4_14.